

Lab 3: Working with Git History, Stashing, and Reverting Commits



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1. Introduction

This lab provides hands-on practice with advanced Git operations used in modern DevOps workflows. It covers handling local and remote commit conflicts, manual merge conflict resolution, managing ignored and tracked files with `.gitignore` and `git rm --cached`, temporarily storing changes with `git stash`, inspecting and checking out past commits, performing soft vs hard resets, amending commits, safely reverting commits on remote branches, force pushing with caution, running Gitea via Docker Compose in GitHub Codespaces, and publishing a portfolio site using GitHub Pages.

2. Objectives

The objectives of this lab are to:

- Handle local and remote commit conflicts using `git pull --no-rebase` and `git pull --rebase`.
- Resolve merge conflicts manually by editing conflict markers.
- Manage ignored files using `.gitignore` and untrack files with `git rm --cached`.
- Temporarily store uncommitted work using `git stash` and restore it with `git stash pop`.
- Inspect commit history and check out specific commits.
- Perform soft and hard resets and understand their differences.
- Amend the most recent commit using `git commit --amend`.
- Safely undo changes on a remote branch using `git revert`.
- Use `git push --force` on non-shared branches.
- Run a self-hosted Gitea server inside GitHub Codespaces using Docker Compose.
- Publish a static portfolio site using GitHub Pages.

3. Tasks and Evidence

Task 1 — Handling Local and Remote Commit Conflicts

A remote edit was made to `README.md` on GitHub, followed by a local edit committed on the same repository. The push was rejected because the remote had newer commits. The conflict was resolved using both `git pull --no-rebase` (merge) and `git pull --rebase` (linear history) approaches.

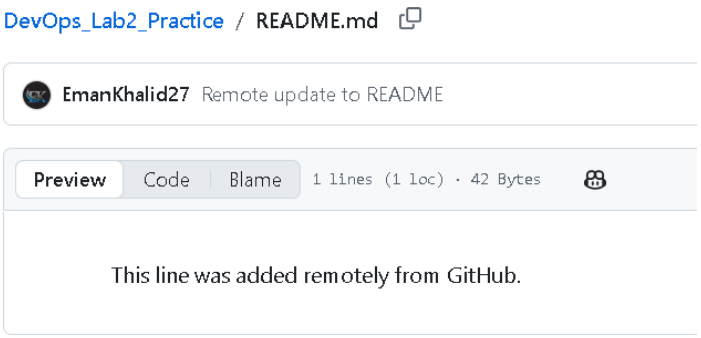


Figure 1.1: Remote edit on GitHub README

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ cat README.md
# Lab02 Git Practice
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ nano README.md
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git add README.md
 README"emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Local update to README"
[main 9d772bd] Local update to README
 1 file changed, 1 insertion(+), 1 deletion(-)
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 1.2: Local commit before pushing — `local_commit.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
To github.com:EmanKhalid27/DevOps_Lab2_Practice.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'github.com:EmanKhalid27/DevOps_Lab2_Practice.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 1.3: Push rejected error — `push_error.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git pull --no-rebase origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 964 bytes | 192.00 KiB/s, done.
From github.com:EmanKhalid27/DevOps_Lab2_Practice
 * branch            main       -> FETCH_HEAD
   71bba01..582309f  main       -> origin/main
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
Automatic merge failed; fix conflicts and then commit the result.
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 1.4: Merge commit from `git pull --no-rebase` — `merge_commit.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Merge remote and local README changes"
[main 253876b] Merge remote and local README changes
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git push -u origin main
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 3 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 666 bytes | 111.00 KiB/s, done.
Total 6 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To github.com:EmanKhalid27/DevOps_Lab2_Practice.git
   582309f..253876b  main -> main
branch 'main' set up to track 'origin/main'.
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 1.5: Successful push after merge — `push_after_merge.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git pull --rebase origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 976 bytes | 195.00 KiB/s, done.
From github.com:EmanKhalid27/DevOps_Lab2_Practice
 * branch            main       -> FETCH_HEAD
   addc0b2..4552b6a  main       -> origin/main
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
error: could not apply c894644... Local update for rebase conflict
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --abort".
hint: Disable this message with "git config set advice.mergeConflict false"
Could not apply c894644... # Local update for rebase conflict
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 1.6: Rebase pull output — `rebase_pull.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git rebase --continue
[detached HEAD 6482779] Local update for rebase conflict
 1 file changed, 2 insertions(+)
Successfully rebased and updated refs/heads/main.
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 3 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 337 bytes | 168.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:EmanKhalid27/DevOps_Lab2_Practice.git
   4552b6a..6482779  main -> main
branch 'main' set up to track 'origin/main'.
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 1.7: Successful push after rebase — `push_after_rebase.png`

Task 2 — Manual Merge Conflict Resolution

The same line in `README.md` was edited both remotely and locally, causing a merge conflict. Conflict markers were manually resolved by keeping both versions.

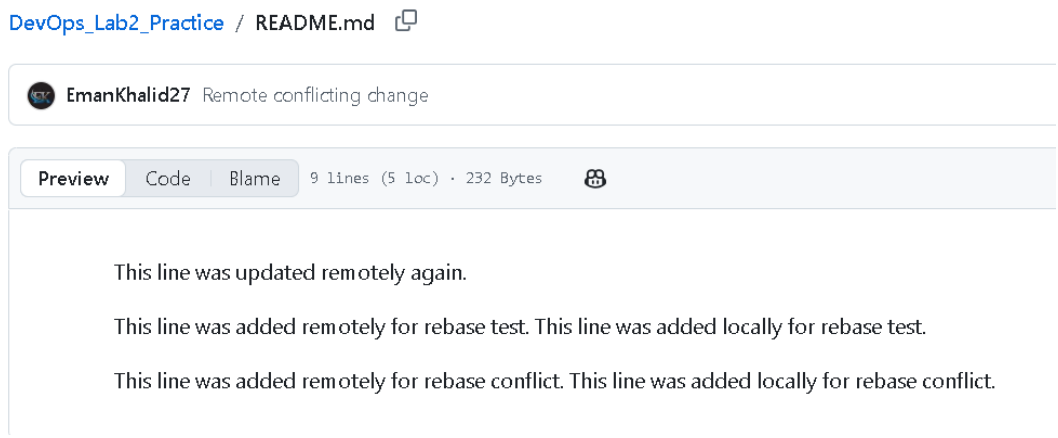


Figure 2.1: Remote conflicting edit — `remote_conflict_edit.png`

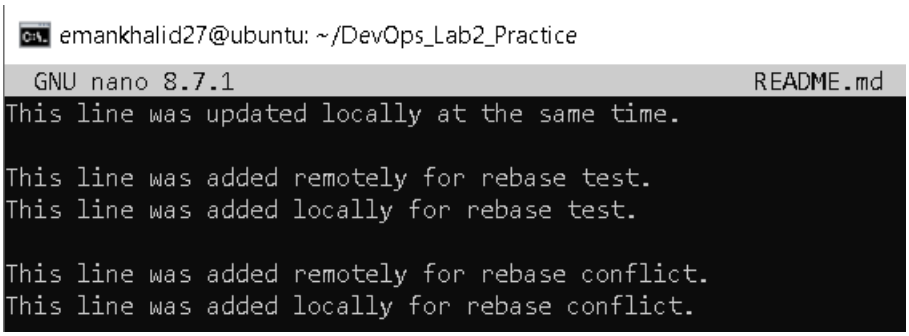


Figure 2.2: Local conflicting edit — `local_conflict_edit.png`

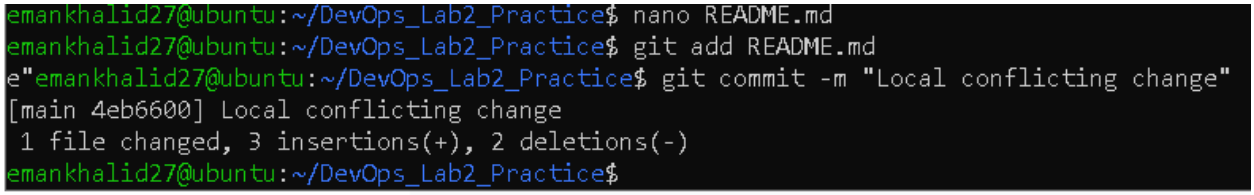


Figure 2.3: Local commit of conflicting change — `local_conflict_commit.png`

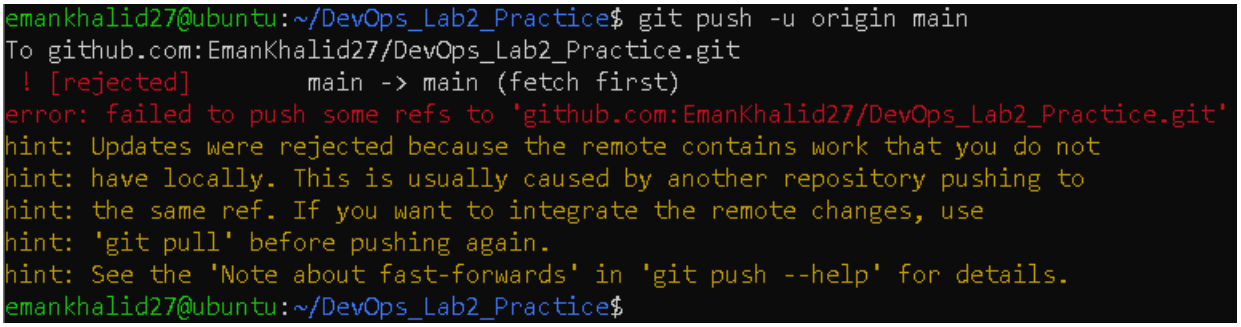


Figure 2.4: Push rejected error — `conflict_push_error.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git pull --rebase origin main
remote: Enumerating objects: 11, done.
remote: Counting objects: 100% (11/11), done.
remote: Compressing objects: 100% (9/9), done.
remote: Total 9 (delta 6), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (9/9), 2.78 KiB | 123.00 KiB/s, done.
From github.com:EmanKhalid27/DevOps_Lab2_Practice
 * branch            main       -> FETCH_HEAD
   6482779..9137d81  main       -> origin/main
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
error: could not apply 4eb6600... Local conflicting change
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --abort".
hint: Disable this message with "git config set advice.mergeConflict false"
Could not apply 4eb6600... # Local conflicting change
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 2.5: Conflict message from `git pull --rebase` — `conflict_message.png`

```
emankhalid27@ubuntu: ~/DevOps_Lab2_Practice
  GNU nano 8.7.1                                        README.md *
This line was updated remotely again.
This line was updated locally at the same time.
```

Figure 2.6: Resolved README file — `resolved_readme.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ nano README.md
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git add README.md
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git rebase --continue
[detached HEAD 373b614] Local conflicting change
 1 file changed, 1 insertion(+), 1 deletion(-)
Successfully rebased and updated refs/heads/main.
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 2.7: Rebase continued successfully — `rebase_continue.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 3 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 354 bytes | 354.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:EmanKhalid27/DevOps_Lab2_Practice.git
   9137d81..373b614  main -> main
branch 'main' set up to track 'origin/main'.
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 2.8: Push after resolving conflict — `push_after_resolve.png`

Task 3 — Managing Ignored Files with .gitignore

A `textfiles/` directory with three files was created and pushed. A `.gitignore` file was then added to ignore `textfiles/*`. The folder remained on GitHub until `git rm -r --cached textfiles` was used to untrack it locally while keeping the files on disk.

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git add .
h three files"
git push origin mainemankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Added textfiles directory with thre
e files"
[main 6488d56] Added textfiles directory with three files
 3 files changed, 3 insertions(+)
 create mode 100644 textfiles/a.txt
 create mode 100644 textfiles/b.txt
 create mode 100644 textfiles/c.txt
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 3 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (6/6), 454 bytes | 90.00 KiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:EmanKhalid27/DevOps_Lab2_Practice.git
   373b614..6488d56  main -> main
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 3.1: Push textfiles directory — `push_textfiles.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ echo "textfiles/*" > .gitignore
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ cat .gitignore
textfiles/*
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git add .gitignore
ignore textfiles directory"
git push origin mainemankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Added .gitignore to ignor
e textfiles directory"
[main 9539682] Added .gitignore to ignore textfiles directory
 1 file changed, 1 insertion(+)
 create mode 100644 .gitignore
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 3 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 310 bytes | 155.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:EmanKhalid27/DevOps_Lab2_Practice.git
   6488d56..9539682  main -> main
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 3.2: Add .gitignore and push — `gitignore_push.png`

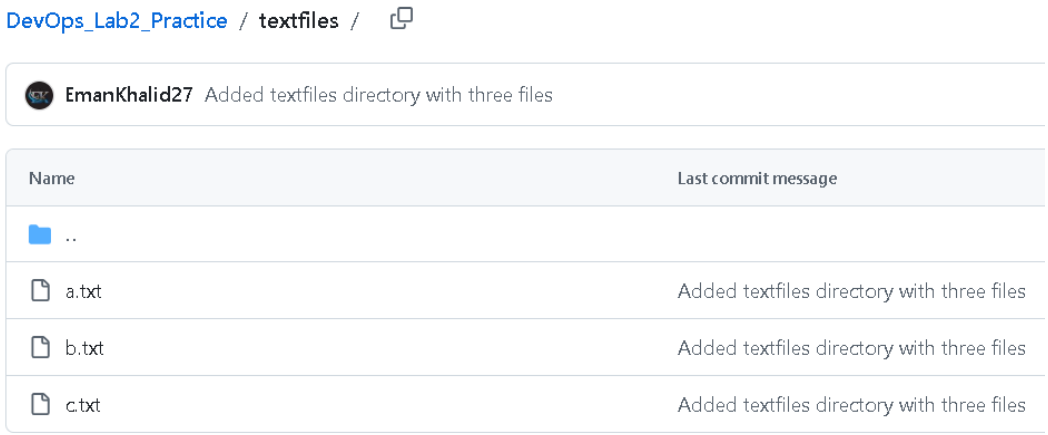


Figure 3.3: Repo still has textfiles — `repo_still_has_textfiles.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git rm -r --cached textfiles
rm 'textfiles/a.txt'
rm 'textfiles/b.txt'
rm 'textfiles/c.txt'
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ ls textfiles/
a.txt  b.txt  c.txt
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git add .
ectory"
git push origin mainemankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Removed tracked textfiles direc
tory"
[main 8f2d473] Removed tracked textfiles directory
 3 files changed, 3 deletions(-)
 delete mode 100644 textfiles/a.txt
 delete mode 100644 textfiles/b.txt
 delete mode 100644 textfiles/c.txt
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 3 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 246 bytes | 123.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:EmanKhalid27/DevOps_Lab2_Practice.git
   9539682..8f2d473  main -> main
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 3.4: Remove tracked files with `git rm --cached` — `rm_cached_push.png`

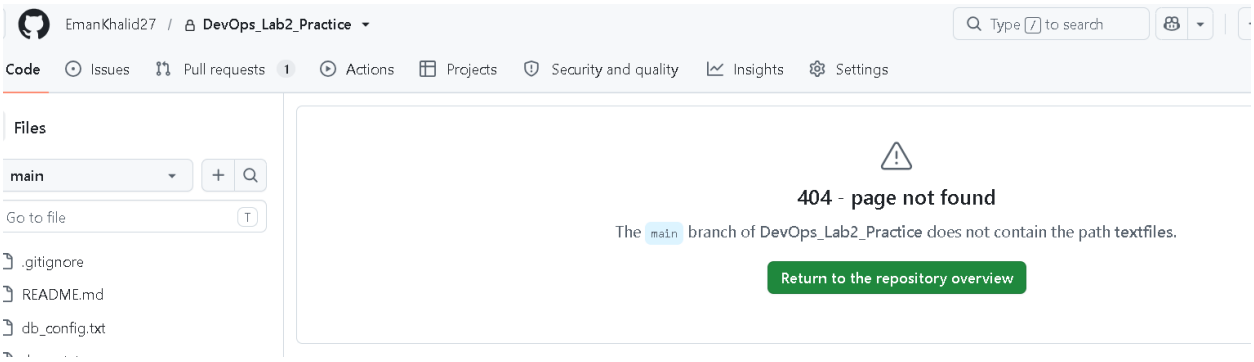


Figure 3.5: Repo without textfiles — `repo_textfiles_removed.png`

Task 4 — Temporary Changes with git stash

Uncommitted changes on `feature-branch` prevented switching branches. `git stash` saved the changes, allowing a clean branch switch. The changes were later restored with `git stash pop`.

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ nano README.md
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git add README.md
ADME.md file"emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Changes the README.md file"
[feature-branch 735851b] Changes the README.md file
 1 file changed, 1 insertion(+)
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ cat README.md
This line was updated remotely again.
This line was updated locally at the same time.

This line was added remotely for rebase test.
This line was added locally for rebase test.

This line was added remotely for rebase conflict.
This line was added locally for rebase conflict.

This line was added in feature-branch.
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 4.1: Modified README on feature-branch — `modified_readme.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ nano README.md
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch feature-branch
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
error: Your local changes to the following files would be overwritten by checkout:
        README.md
Please commit your changes or stash them before you switch branches.
Aborting
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 4.2: Checkout error due to uncommitted changes — `checkout_error.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git stash
Saved working directory and index state WIP on feature-branch: 735851b Changes the README.md file
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 4.3: Stash command executed — `stash_command.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 4.4: Successfully switched to main — `branch_switched.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git checkout feature-branch
Switched to branch 'feature-branch'
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 4.5: Switched back to feature-branch — `back_to_feature.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch feature-branch
nothing to commit, working tree clean
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 4.6: Clean working tree — `status_clean.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git stash pop
On branch feature-branch
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
Dropped refs/stash@{0} (44e24c7eafc94e7e13d324555461b6586ca47ab8)
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 4.7: Stash popped successfully — `stash_pop.png`

Task 5 — Checkout a Specific Commit

The commit history was inspected with `git log --oneline`, and an older commit was checked out, putting the repository into a "detached HEAD" state. The repository was returned to `main` afterward.

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline
8f2d473 (HEAD -> main, origin/main) Removed tracked textfiles directory
9539682 Added .gitignore to ignore textfiles directory
6488d56 Added textfiles directory with three files
373b614 Local conflicting change
9137d81 Remote conflicting change
cbf1479 Remote conflicting change
68f3fe4 Remote conflicting change
6482779 Local update for rebase conflict
4552b6a Remote update for rebase conflict
addc0b2 Local update for rebase test
6730bfc Remote update for rebase test
0ee900f Local update for rebase conflict
228e6c6 Local update for rebase test
48fb914 Remote update for rebase test
3f54c69 Remote update for rebase test
253876b Merge remote and local README changes
9d772bd Local update to README
582309f Remote update to README
```

Figure 5.1: Commit log before checkout — `log_before_checkout.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git checkout 6488d56
Note: switching to '6488d56'.

You are in 'detached HEAD' state. You can look around, make experimental
changes and commit them, and you can discard any commits you make in this
state without impacting any branches by switching back to a branch.

If you want to create a new branch to retain commits you create, you may
do so (now or later) by using -c with the switch command. Example:

  git switch -c <new-branch-name>

Or undo this operation with:

  git switch -

Turn off this advice by setting config variable advice.detachedHead to false

HEAD is now at 6488d56 Added textfiles directory with three files
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 5.2: Detached HEAD state — `detached_head.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Previous HEAD position was 6488d56 Added textfiles directory with three files
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
```

Figure 5.3: Back to main branch — `back_to_main.png`

Task 6 — Soft vs Hard Reset

Two test commits were made, and both soft and hard resets were performed with verification of the file contents and commit history at each stage.

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ nano README.md
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git add .
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Added test line"
[main 2bc19e5] Added test line
 1 file changed, 2 insertions(+)
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 6.1: First commit — `first_commit.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ nano README.md
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git add .
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Second test commit"
[main 5cee500] Second test commit
 1 file changed, 1 insertion(+)
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 6.2: Second commit — `second_commit.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline
5cee500 (HEAD -> main) Second test commit
2bc19e5 Added test line
8f2d473 (origin/main) Removed tracked textfiles directory
9539682 Added .gitignore to ignore textfiles directory
6488d56 Added textfiles directory with three files
373b614 Local conflicting change
9137d81 Remote conflicting change
cbf1479 Remote conflicting change
68f3fe4 Remote conflicting change
6482779 Local update for rebase conflict
4552b6a Remote update for rebase conflict
addc0b2 Local update for rebase test
6730bfc Remote update for rebase test
0ee900f Local update for rebase conflict
228e6c6 Local update for rebase test
48fb914 Remote update for rebase test
3f54c69 Remote update for rebase test
253876b Merge remote and local README changes
9d772bd Local update to README
582309f Remote update to README
```

Figure 6.3: Log before reset — `log_before_reset.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ cat README.md
This is the first test line for reset.
This is the second test line for reset.

This line was updated remotely again.
This line was updated locally at the same time.

This line was added remotely for rebase test.
This line was added locally for rebase test.

This line was added remotely for rebase conflict.
This line was added locally for rebase conflict.

emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 6.4: File before reset (both lines present) — `file_before_reset.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git reset --soft HEAD~1
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 6.5: Soft reset performed — `soft_reset.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline
2bc19e5 (HEAD -> main) Added test line
8f2d473 (origin/main) Removed tracked textfiles directory
9539682 Added .gitignore to ignore textfiles directory
6488d56 Added textfiles directory with three files
```

Figure 6.6: Log after soft reset — `log_after_soft_reset.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ cat README.md
This is the first test line for reset.
This is the second test line for reset.

This line was updated remotely again.
This line was updated locally at the same time.

This line was added remotely for rebase test.
This line was added locally for rebase test.

This line was added remotely for rebase conflict.
This line was added locally for rebase conflict.

emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 6.7: File after soft reset (changes kept) — `file_after_soft_reset.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ cat README.md
This is the first test line for reset.

This line was updated remotely again.
This line was updated locally at the same time.

This line was added remotely for rebase test.
This line was added locally for rebase test.

This line was added remotely for rebase conflict.
This line was added locally for rebase conflict.

emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 6.8: Recommit after soft reset — `file_after_hard_reset.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git reset --hard HEAD~1
HEAD is now at 2bc19e5 Added test line
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 6.9: Hard reset performed — `hard_reset.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline
2bc19e5 (HEAD -> main) Added test line
8f2d473 (origin/main) Removed tracked textfiles directory
9539682 Added .gitignore to ignore textfiles directory
6488d56 Added textfiles directory with three files
```

Figure 6.10: Log after hard reset — `log_after_hard_reset.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ cat README.md
This is the first test line for reset.

This line was updated remotely again.
This line was updated locally at the same time.

This line was added remotely for rebase test.
This line was added locally for rebase test.

This line was added remotely for rebase conflict.
This line was added locally for rebase conflict.

emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 6.11: File after hard reset (changes discarded) — `file_after_hard_reset.png`

Task 7 — Amending the Last Commit

A commit was made and then amended using `git commit --amend` to include an additional file and a new commit message.

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ nano README.md
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git add .
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Fix log message"
[main 088c361] Fix log message
 1 file changed, 2 insertions(+)
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 7.1: First amend commit — `first_amend_commit.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ nano README.md
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git add .
al line"emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git commit --amend -m "Fix log message with additional
 line"
[main 666b301] Fix log message with additional line
 Date: Sun Oct 11 07:16:22 2026 +0000
 1 file changed, 3 insertions(+)
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 7.2: Amended commit output — `amend_commit.png`

Task 8 — Reverting a Commit

A temporary file was committed and pushed. It was then safely reverted using `git revert`, which created a new commit that undid the changes without rewriting history.

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ echo "temporary text" >> temp.txt
Added temporary text file"
git push origin mainemankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git add .
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Added temporary text file"
[main 9ec8afc] Added temporary text file
 1 file changed, 1 insertion(+)
 create mode 100644 temp.txt
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 3 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (9/9), 877 bytes | 292.00 KiB/s, done.
Total 9 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (5/5), completed with 2 local objects.
To github.com:EmanKhalid27/DevOps_Lab2_Practice.git
   8f2d473..9ec8afc  main -> main
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 8.1: Commit temp file — `commit_temp_file.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline -3
9ec8afc (HEAD -> main, origin/main) Added temporary text file
666b301 Fix log message with additional line
2bc19e5 Added test line
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git revert 9ec8afc
[main 164139f] Revert "Added temporary text file"
 1 file changed, 1 deletion(-)
 delete mode 100644 temp.txt
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 8.2: Revert commit created — `revert_commit.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 3 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 263 bytes | 263.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:EmanKhalid27/DevOps_Lab2_Practice.git
   9ec8afc..164139f  main -> main
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 8.3: Revert commit pushed — `revert_push.png`

Task 9 — Force Push

A new branch `test-force` was created, a commit made, and pushed. After a local hard reset, a normal push was rejected. A force push was then used to overwrite the remote branch history.

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git checkout -b test-force
Switched to a new branch 'test-force'
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 9.1: New branch test-force — `new_branch.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ nano README.md
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git add .
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Force push test commit"
[test-force 6fdceb1] Force push test commit
 1 file changed, 1 insertion(+)
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 9.2: Force commit — `force_commit.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git push origin test-force
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 3 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 317 bytes | 105.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
remote:
remote: Create a pull request for 'test-force' on GitHub by visiting:
remote:      https://github.com/EmanKhalid27/DevOps_Lab2_Practice/pull/new/test-force
remote:
To github.com:EmanKhalid27/DevOps_Lab2_Practice.git
 * [new branch]      test-force -> test-force
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 9.3: Push force branch — `push_force_branch.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git reset --hard HEAD~1
HEAD is now at 164139f Revert "Added temporary text file"
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 9.4: Hard reset — `hard_reset_force.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git push origin test-force
To github.com:EmanKhalid27/DevOps_Lab2_Practice.git
 ! [rejected]        test-force -> test-force (non-fast-forward)
error: failed to push some refs to 'github.com:EmanKhalid27/DevOps_Lab2_Practice.git'
hint: Updates were rejected because the tip of your current branch is behind
hint: its remote counterpart. If you want to integrate the remote changes,
hint: use 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 9.5: Normal push rejected — `normal_push.png`

```
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$ git push origin test-force --force
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:EmanKhalid27/DevOps_Lab2_Practice.git
 + 6fdceb1...164139f test-force -> test-force (forced update)
emankhalid27@ubuntu:~/DevOps_Lab2_Practice$
```

Figure 9.6: Force push output — `force_push.png`

Task 10 — Running Gitea in GitHub Codespaces

The Gitea repository was forked, opened in GitHub Codespaces, and Gitea was started using Docker Compose. Port 3000 was forwarded publicly, and Gitea was installed with an admin account. A new repository was then created in Gitea.

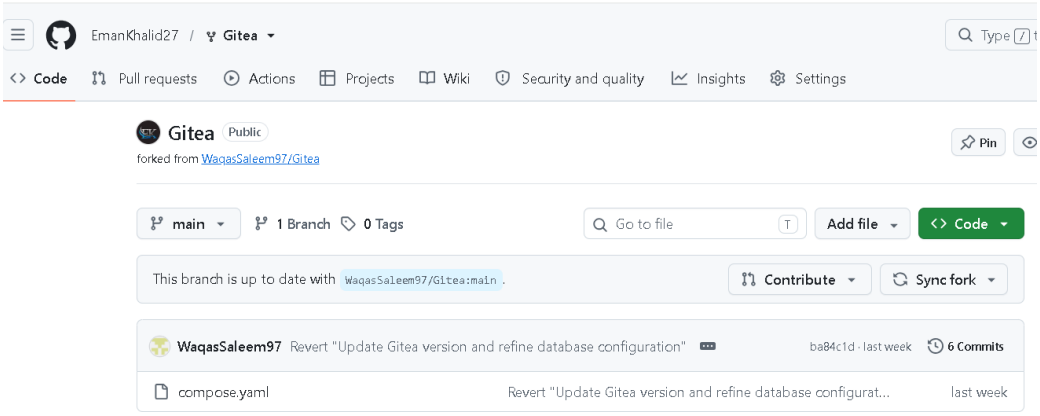
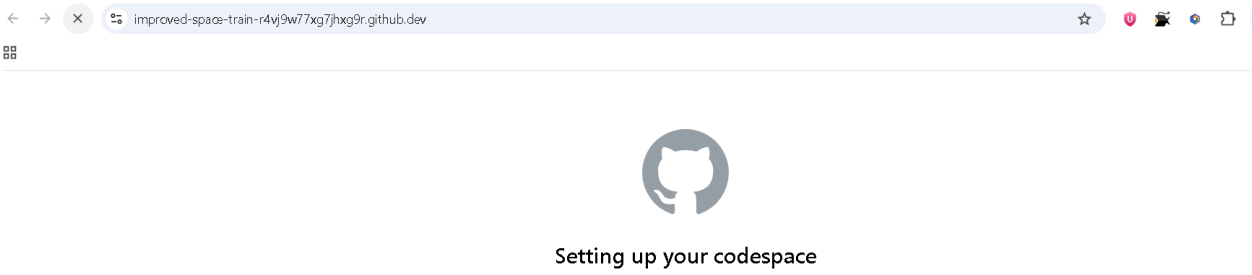


Figure 10.1: Forked Gitea repository — `forked_gitea.png`



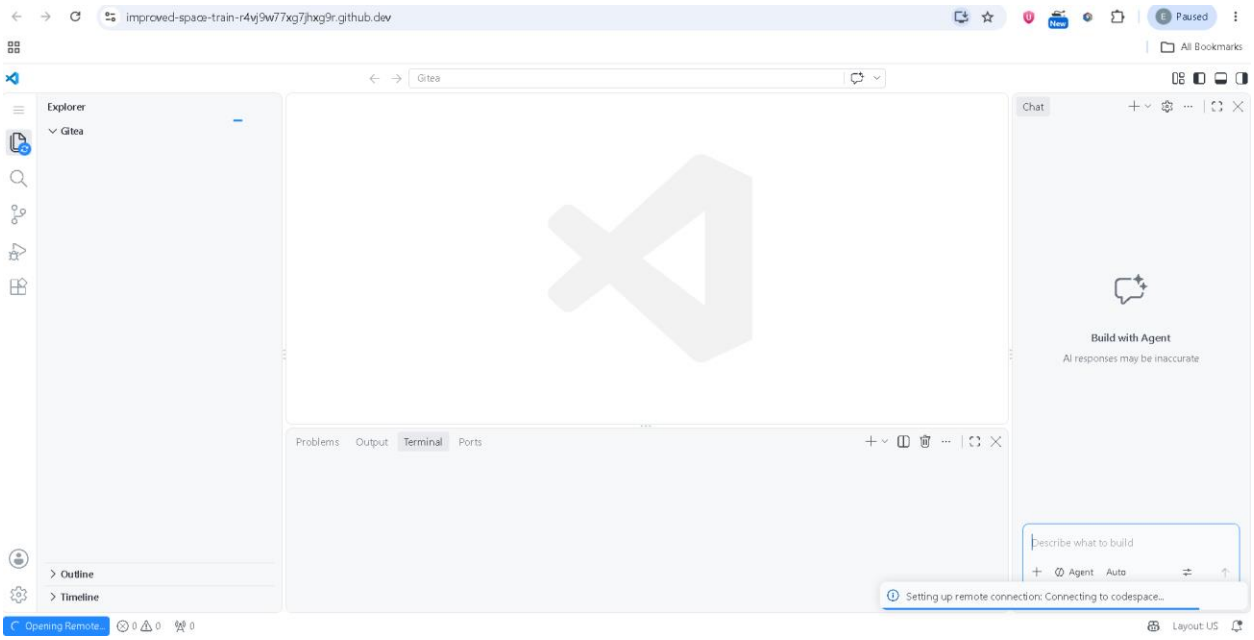
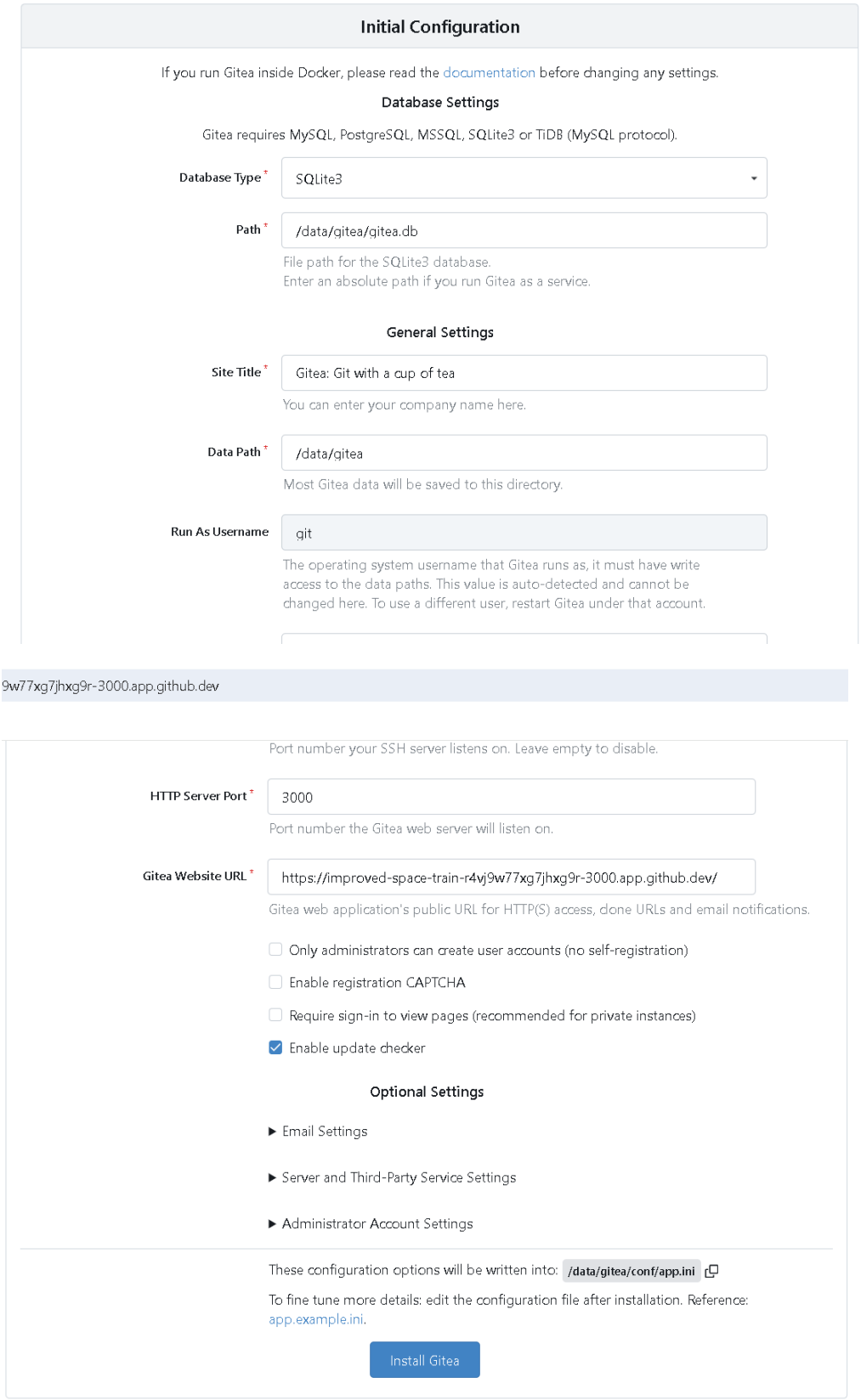


Figure 10.2: Codespace loading — `codespace_loading.png`

```
@EmanKhalid27 ➜ /workspaces/Gitea (main) $ docker compose up -d
[+] up 25/25
 ✓ Image postgres:17-alpine Pulled                                    13.5s
 ✓ Image gitea/gitea:latest Pulled                                     9.8s
 ✓ Network webnet           Created                                    0.1s
 ✓ Volume gitea_postgres    Created                                    0.0s
 ✓ Volume gitea             Created                                    0.0s
 ✓ Container gitea          Started                                    8.3s
 ✓ Container gitea_db       Started                                    7.9s
@EmanKhalid27 ➜ /workspaces/Gitea (main) $ docker ps
CONTAINER ID   IMAGE                COMMAND                  CREATED          STATUS          PORTS
                                                NAMES
2c9ee3d182a0   postgres:17-alpine   "docker-entrypoint.s…"   19 seconds ago   Up 12 seconds   5432/tcp
                                                gitea_db
75fb27e48dd5   gitea/gitea:latest   "/usr/bin/entrypoint…"   19 seconds ago   Up 11 seconds   22/tcp, 0.0.
0.0:3000->3000/tcp, [::]:3000->3000/tcp   gitea
@EmanKhalid27 ➜ /workspaces/Gitea (main) $
```

Figure 10.3: Docker containers running — `docker_up.png`



9w77xg7jhxg9r-3000.app.github.dev

Figure 10.4: Gitea install page — `gitea_install_page.png`

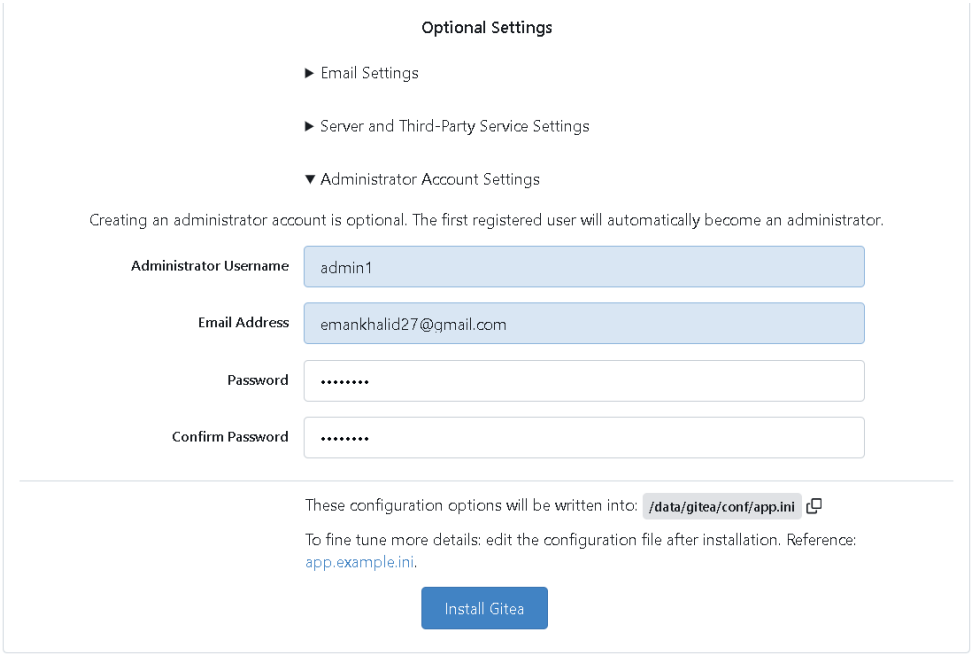


Figure 10.5: Admin account setup — `admin_setup.png`

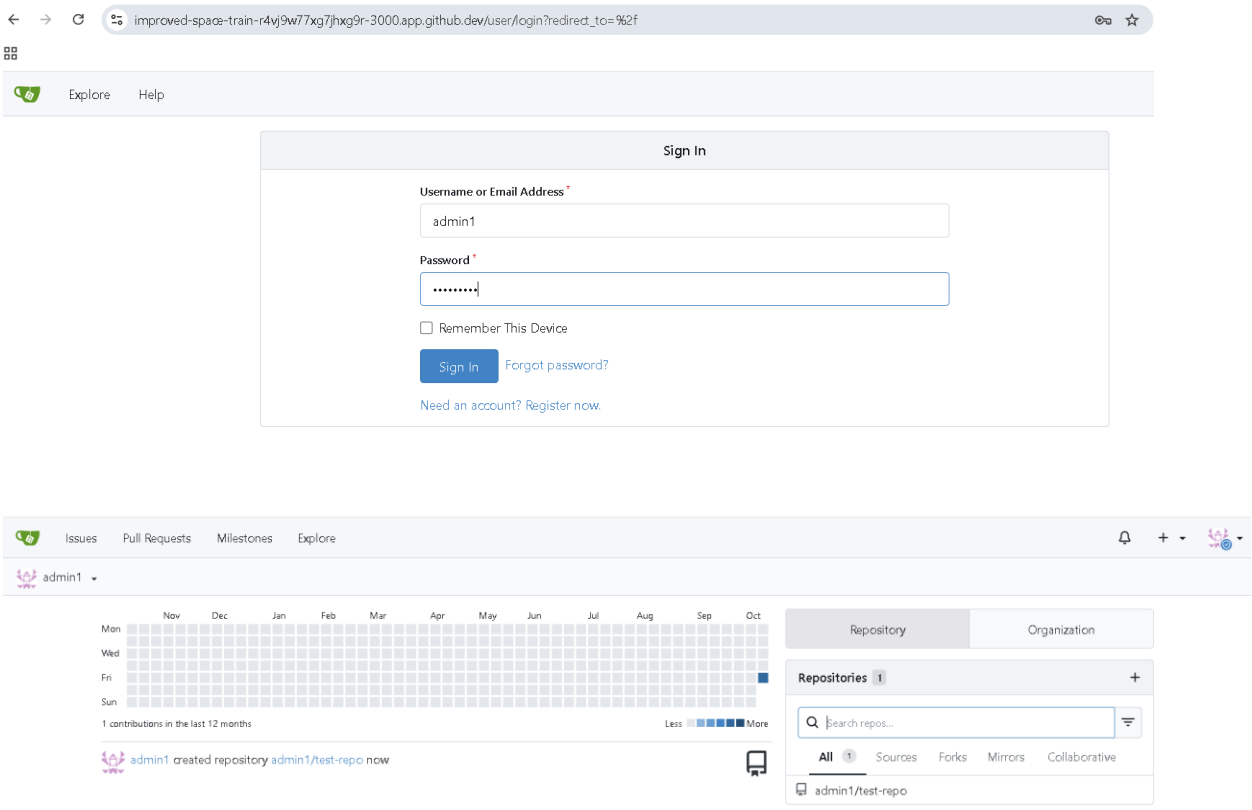


Figure 10.6: Gitea dashboard after login — `gitea_dashboard.png`

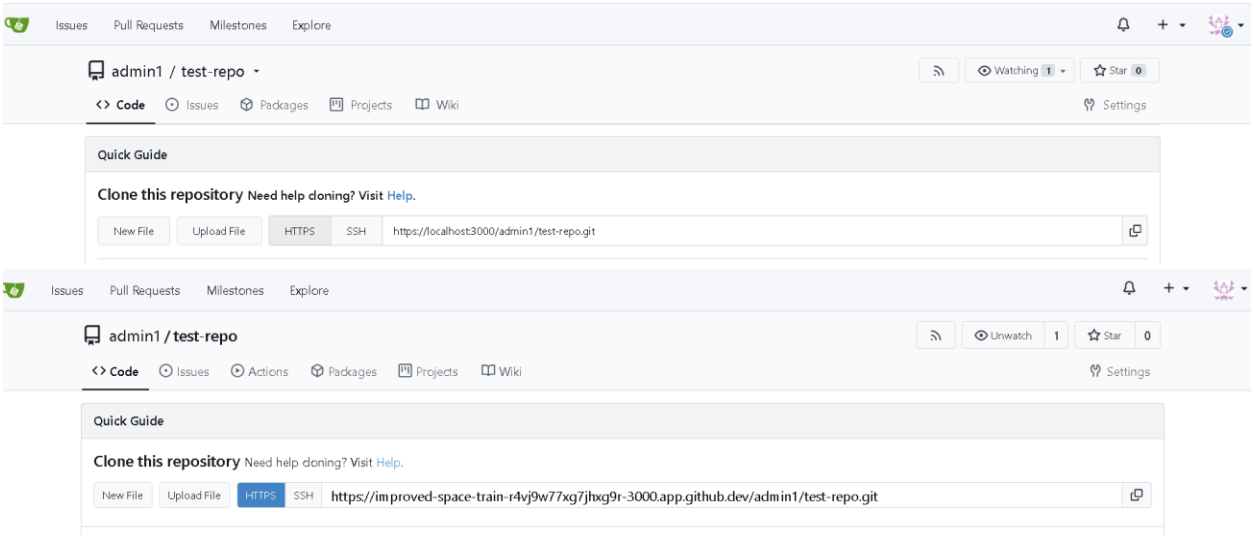


Figure 10.7: New repository in Gitea — `gitea_new_repo.png`

Task 11 — GitHub Pages Portfolio Site

A public repository named `EmanKhalid27.github.io` was created. A static portfolio site was pushed to it, GitHub Pages was enabled from the `main` branch, and the site was confirmed live.

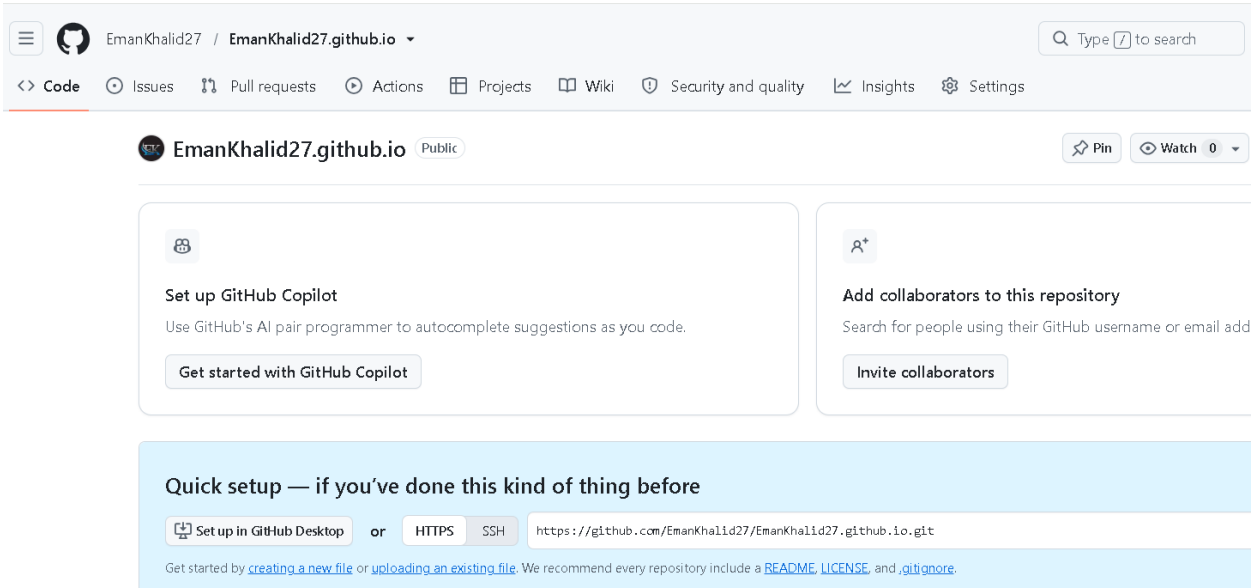


Figure 11.1: GitHub Pages repository — `github_pages_repo.png`

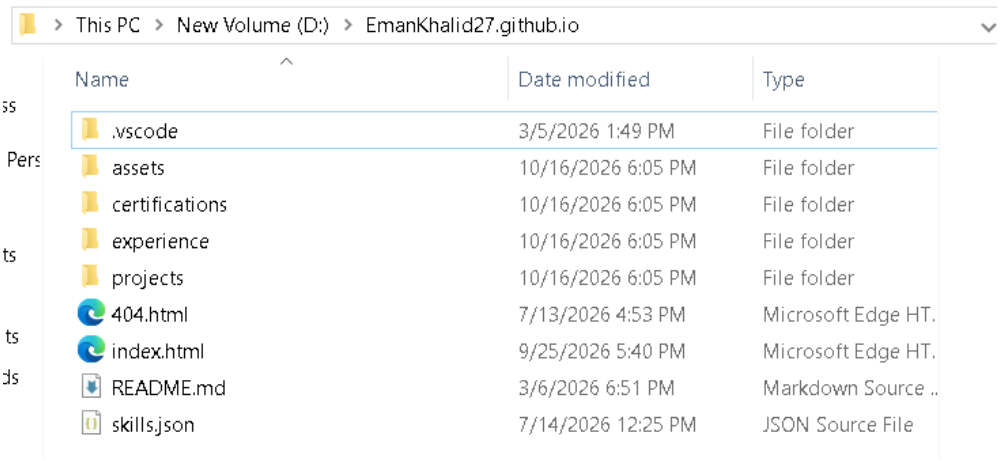


Figure 11.2: Local static site files — `local_static_site.png`

```
D:\EmanKhalid27.github.io>git push -u origin main
Enumerating objects: 71, done.
Counting objects: 100% (71/71), done.
Delta compression using up to 8 threads
Compressing objects: 100% (68/68), done.
Writing objects: 100% (71/71), 15.39 MiB | 829.00 KiB/s, done.
Total 71 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (5/5), done.
To github.com:EmanKhalid27/EmanKhalid27.github.io.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

D:\EmanKhalid27.github.io>
```

Figure 11.3: Push static site — `push_static_site.png`

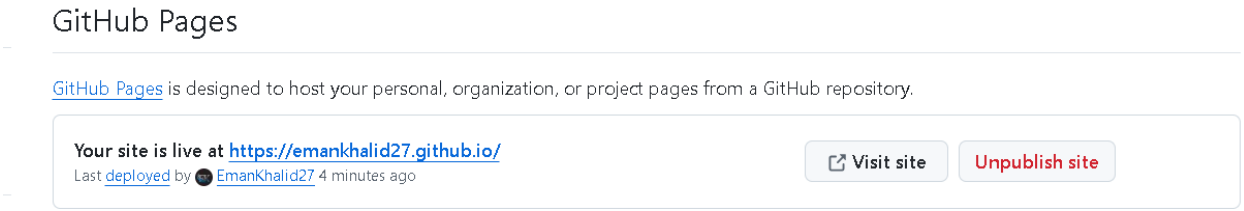


Figure 11.4: GitHub Pages settings — `github_pages_settings.png`

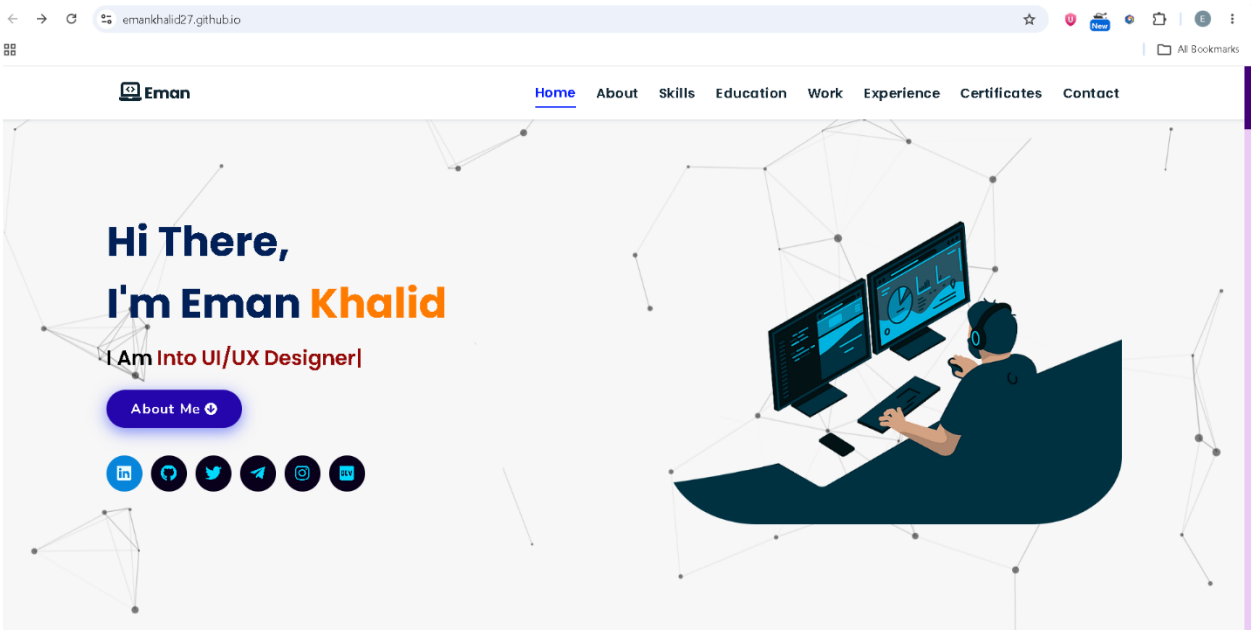


Figure 11.5: Live portfolio site — `live_site.png`

5. Conclusion

This lab provided comprehensive hands-on practice with advanced Git and GitHub workflows. Key concepts covered include conflict resolution, `.gitignore` and `git rm --cached`, `git stash`, detached HEAD state, soft vs hard resets, amending and reverting commits, force pushing, running Gitea in Codespaces via Docker Compose, and publishing a portfolio site using GitHub Pages.